Technical Characterization Sheet

Institute:	Niels Bohr Institute

Product name:	Single Quantum R12
Product serial number:	SQ341
System CE serial number:	040532
Product description:	10-channel superconducting nanowire single photon detection system

Cold head model:	Sumitomo RDK-101D
Compressor model:	Sumitomo F-20L
Electronic driver:	Single Quantum Retina 1200 r1-673

Characterization date:	12/11/2025
Characterized by:	Jordy Stad
Approved by:	Quinten Roest

Cryostat operation

Base temperature (K)	Cooldown time (Hr)
2.1	4

SINGLE QUANTUM
Excellence in photon detection

Specification sheet

Channel	Driver channel	SDE at λ nm (%)[1]		DCR (cps)[2]	Timing jitter (ps)[3]	Critical current (μA)	Operating current (μA)	Output pulse amplitude (mV)	Driver trigger level (mV)
		1309	1545						
1	1	92	-	<5	13	25.6	23.8	1000	100
2	2	91	-	<5	13	26.1	24.3	1050	100
3	3	93	-	<5	14	26.2	24.2	1060	100
4	4	90	-	<5	14	25.7	24.0	1040	100
5	5	95	-	<5	15	28.8	27.1	1180	100
6	6	93	-	<5	13	26.2	24.2	1030	100
8	8	-	91	<10	13	28.1	26.6	1120	100
9	9	-	91	<20	13	28.6	27.0	1160	100
10	10	-	94	<20	15	28.6	27.1	1130	100
12	12	-	93	<20	13	27.4	25.9	1100	100

[1] System detection efficiency at operating current. Measurement uncertainty ± 3%.

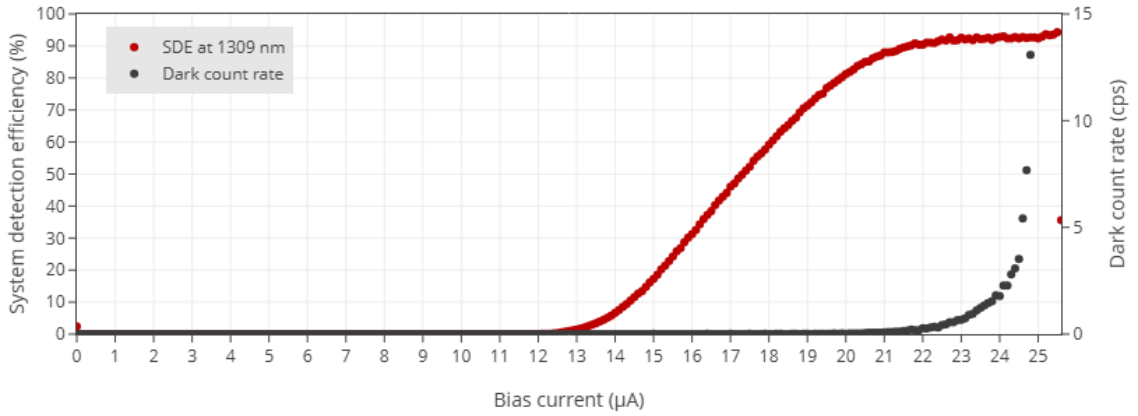
[2] On the next pages, the SDE and DCR are plotted as a function of the bias current.

[3] FWHM value, measurement uncertainty: ± 2 ps. Jitter can be lowered at the expense of higher DCR, by increasing the bias current. More information is given in the next pages.

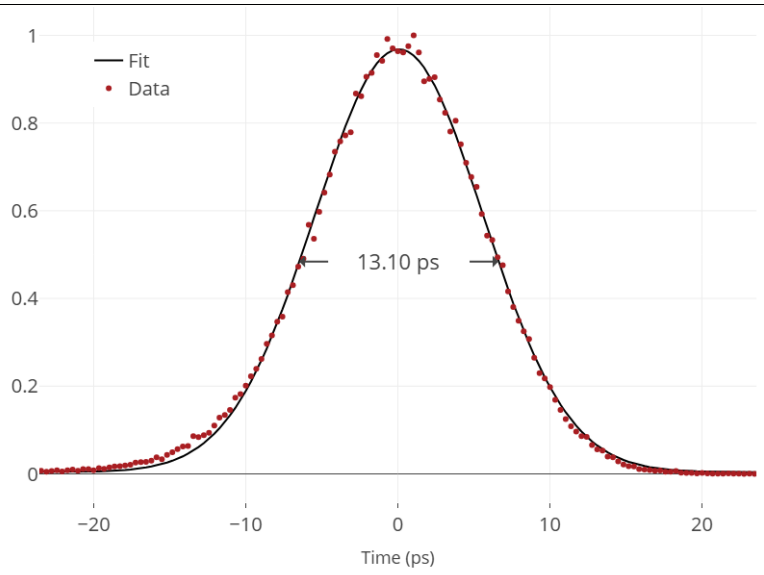
Channel 1

Fiber type: SMF28U-APC
Driver: Channel 1

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
92	<5	13	25.6	23.8	1000	100



SDE and DCR as a function of the bias current

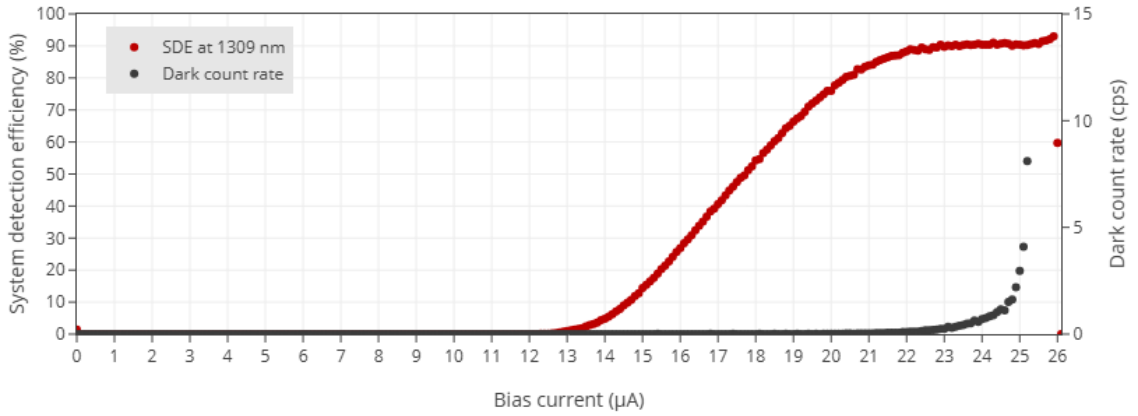


Jitter distribution

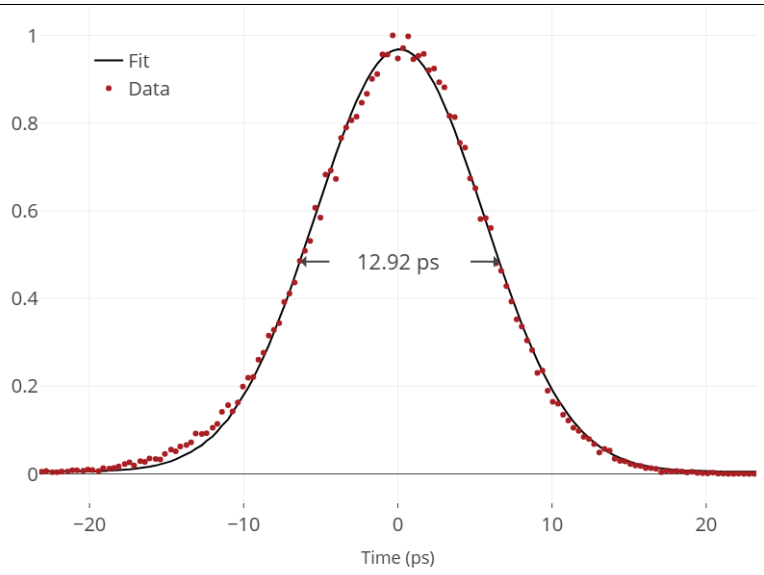
Channel 2

Fiber type: SMF28U-APC
Driver: Channel 2

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
91	<5	13	26.1	24.3	1050	100



SDE and DCR as a function of the bias current

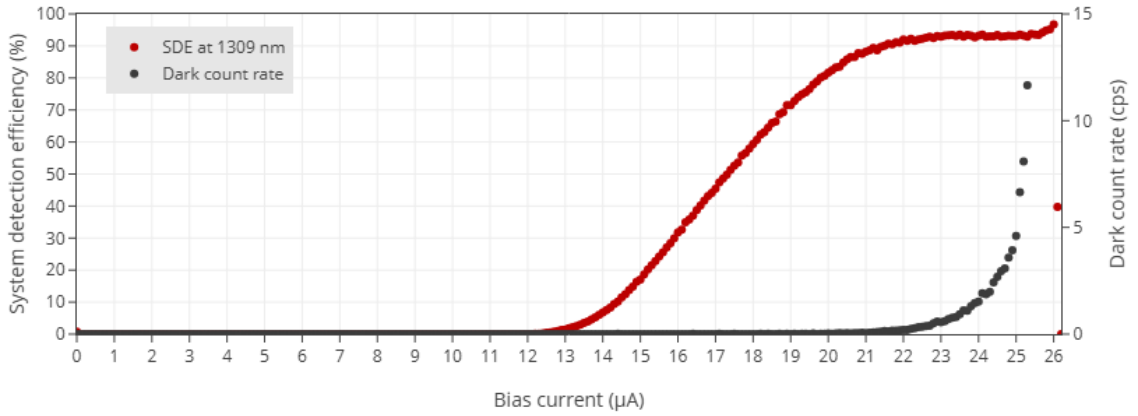


Jitter distribution

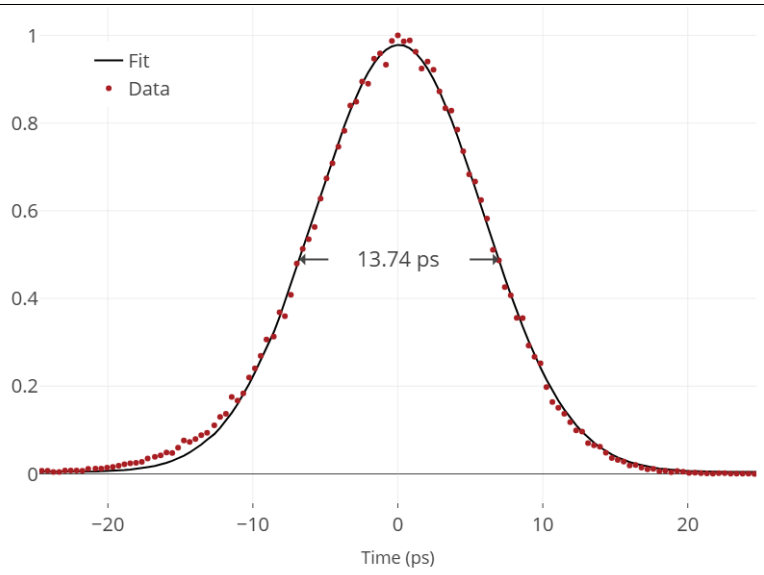
Channel 3

Fiber type: SMF28U-APC
Driver: Channel 3

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
93	<5	14	26.2	24.2	1060	100



SDE and DCR as a function of the bias current

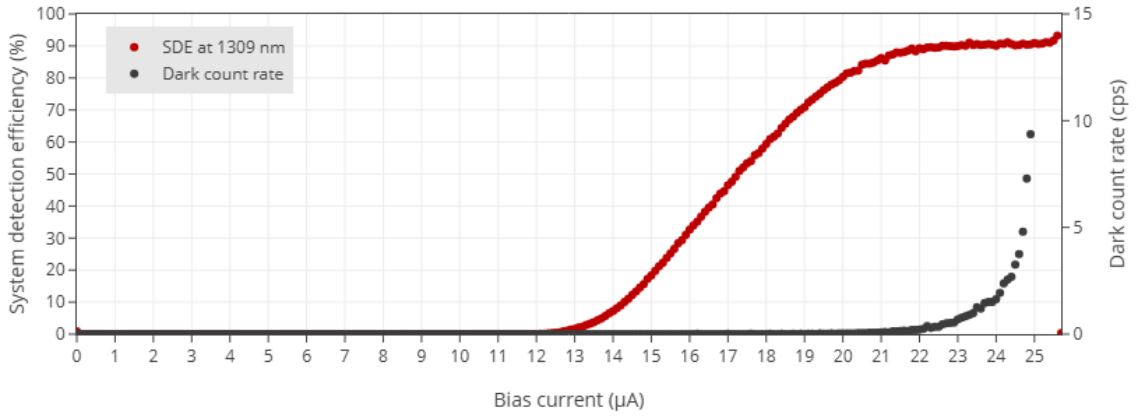


Jitter distribution

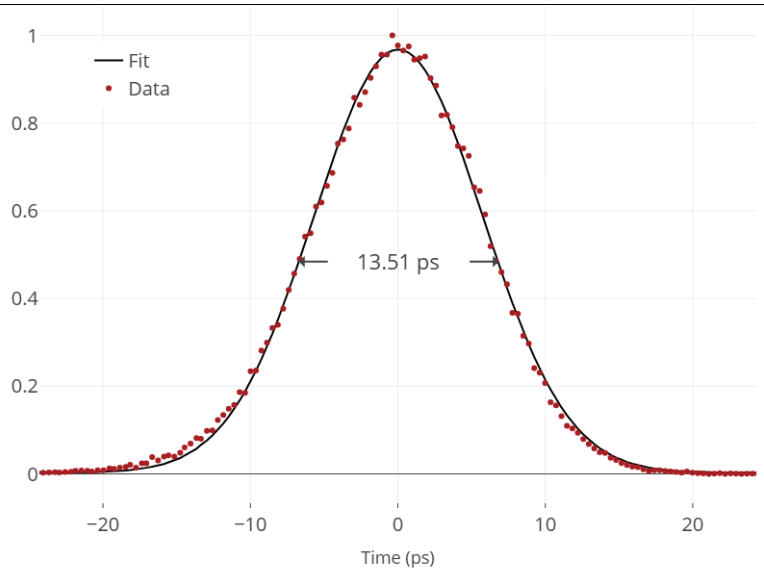
Channel 4

Fiber type: SMF28U-APC
Driver: Channel 4

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
90	<5	14	25.7	24.0	1040	100



SDE and DCR as a function of the bias current

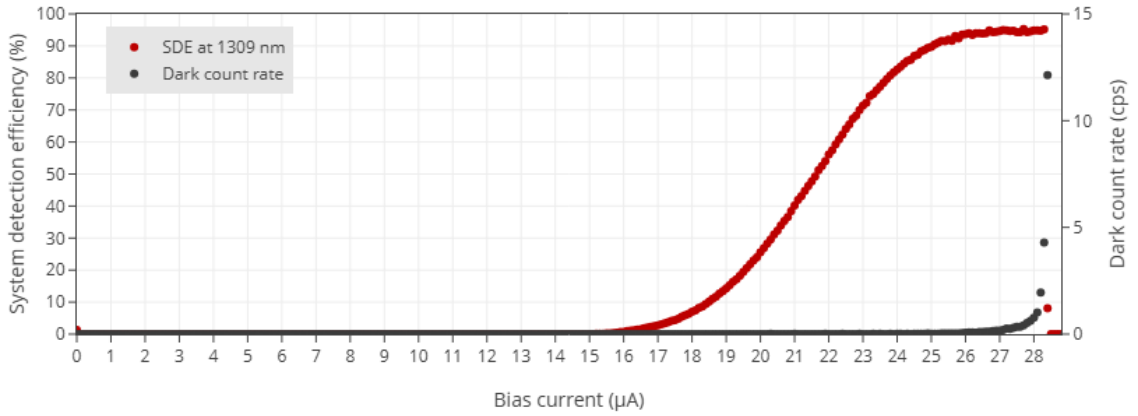


Jitter distribution

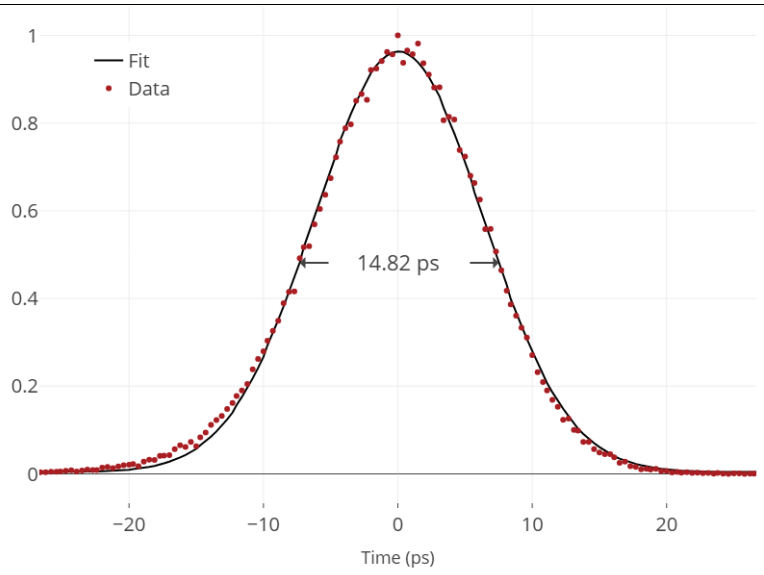
Channel 5

Fiber type: SMF28U-APC
Driver: Channel 5

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
95	<5	15	28.8	27.1	1180	100



SDE and DCR as a function of the bias current

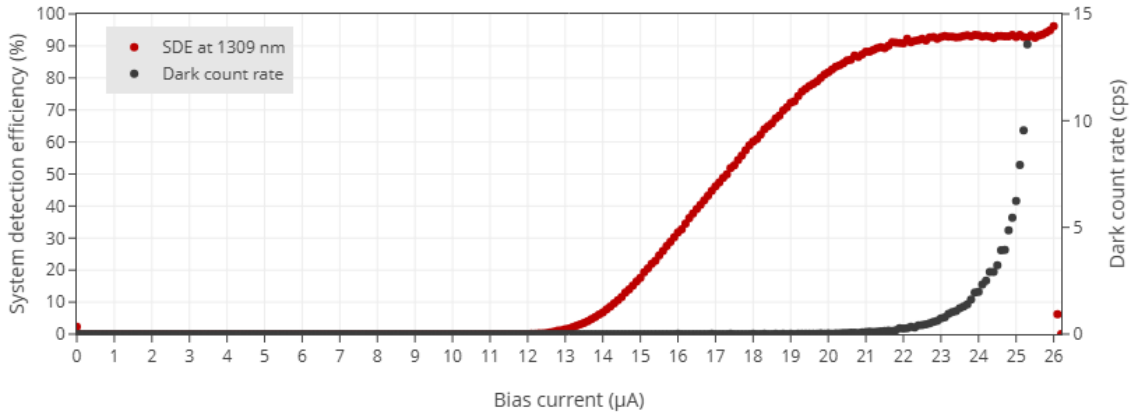


Jitter distribution

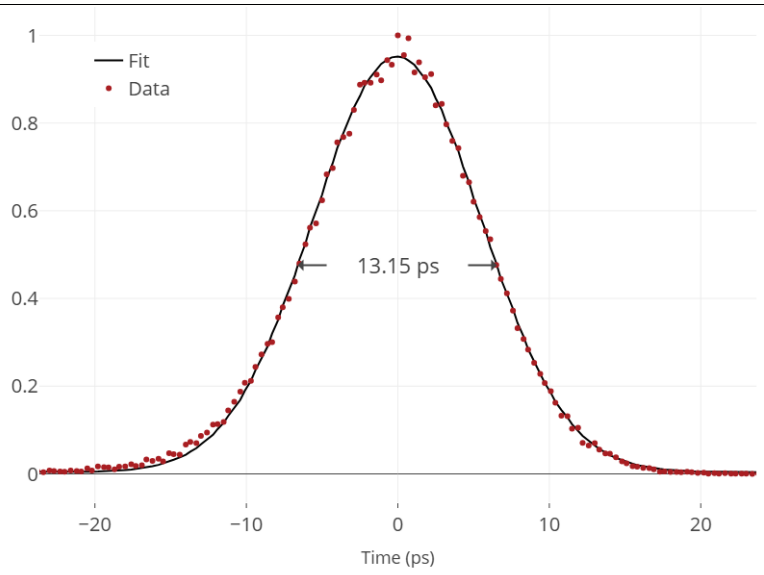
Channel 6

Fiber type: SMF28U-APC
Driver: Channel 6

SDE at 1309 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
93	<5	13	26.2	24.2	1030	100



SDE and DCR as a function of the bias current

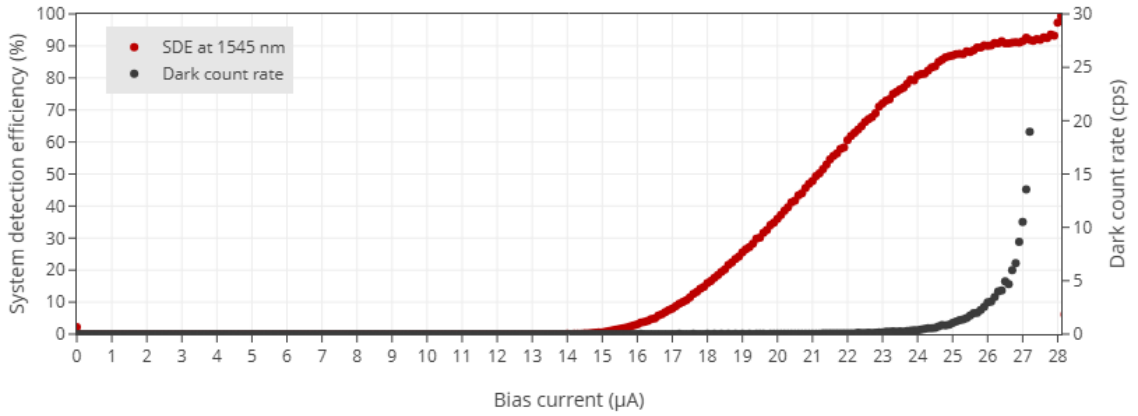


Jitter distribution

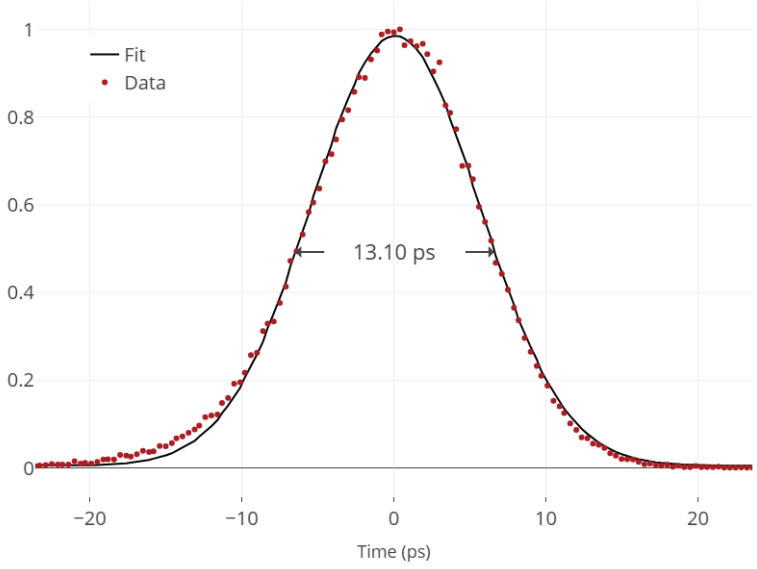
Channel 8

Fiber type: SMF28U-APC
Driver: Channel 8

SDE at 1545 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
91	<10	13	28.1	26.6	1120	100



SDE and DCR as a function of the bias current

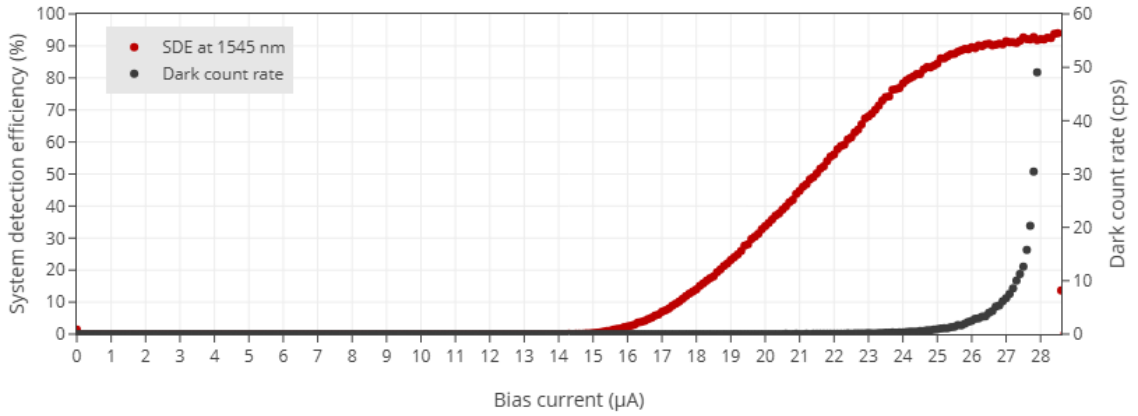


Jitter distribution

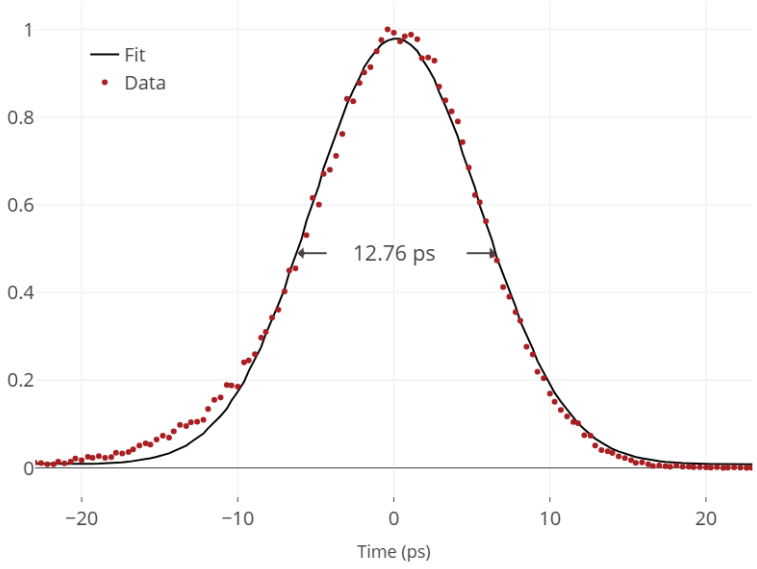
Channel 9

Fiber type: SMF28U-APC
Driver: Channel 9

SDE at 1545 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
91	<20	13	28.6	27.0	1160	100



SDE and DCR as a function of the bias current

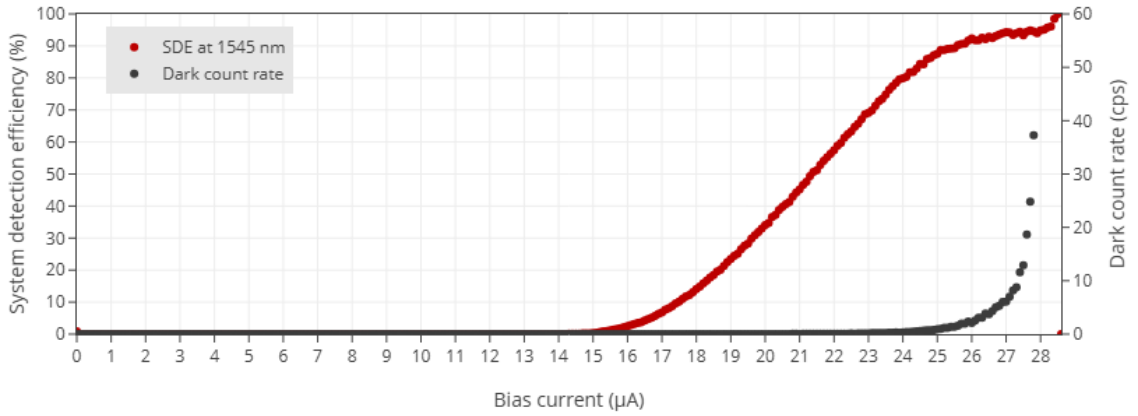


Jitter distribution

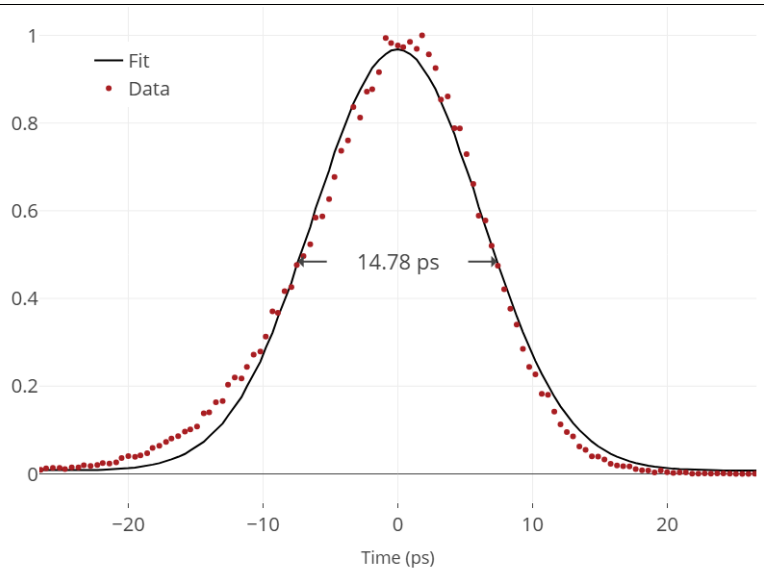
Channel 10

Fiber type: SMF28U-APC
Driver: Channel 10

SDE at 1545 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
94	<20	15	28.6	27.1	1130	100



SDE and DCR as a function of the bias current

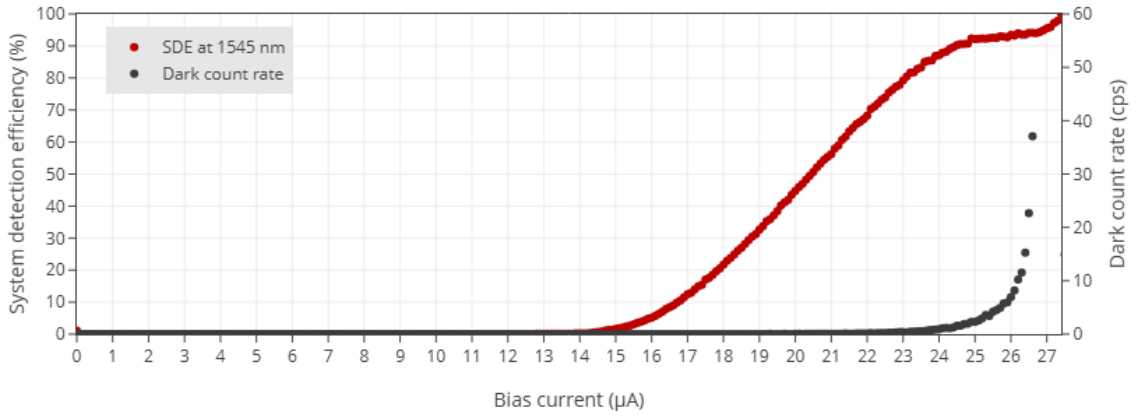


Jitter distribution

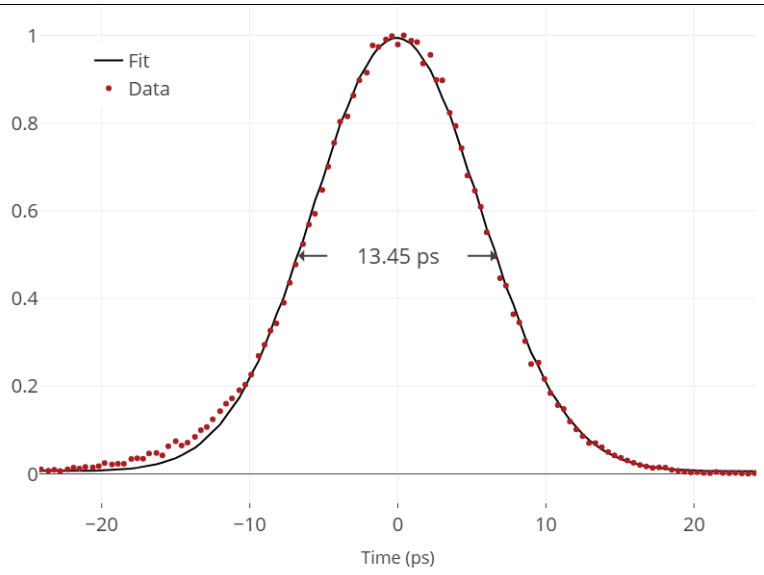
Channel 12

Fiber type: SMF28U-APC
Driver: Channel 12

SDE at 1545 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)	Driver trigger level (mV)
93	<20	13	27.4	25.9	1100	100



SDE and DCR as a function of the bias current



Jitter distribution

Appendix 1: Characterization methods

Dark count rate

The dark count rate (DCR) was measured by sweeping the bias current of the detectors with a step size of 0.1 µA and an integration time of 60 s. During this measurement, all the fiber ports in the front panel of the system were closed, and environmental light was minimized. The reported DCR value for a detector is at a fixed biasing current which is defined as the operating current.

Efficiency

The efficiency experiments were done by coupling a laser light source to the system with wavelengths of either 1309 nm or 1545 nm. The input power towards the system was 3.0 nW + 50 dB attenuation or 2.0 nW + 50 dB attenuation for the 1309 nm or 1545 nm light source, respectively. This can be calculated as an input photon flux of 198k or 156k photons per second, respectively.

With the light source coupled to the system, the bias current of the detectors was swept with a step size of 0.1 µA and an integration time of 600 ms. The trigger level in the driver interface is typically set at 100 – 200 mV.

With the known input photon flux, the bias current dependent system detection efficiency (*SDE*) can be calculated as:

$$SDE = \frac{CR - DCR}{\phi},$$

Where CR is the measured count rate, DCR is the measured dark count rate, and ϕ is the input photon flux. The reported SDE value for a detector is at a fixed operating current.

Timing Jitter

The jitter experiments were done by a pulsed laser light source (Fianium FM-1060) with a built-in photodiode, and an oscilloscope (Lecroy, 4 GHz bandwidth and 40 GS/s sampling). Light pulses of <5 ps were generated with a repetition rate of 50 MHz. The laser light was fiber-coupled to the cryostat and the optical input power was attenuated until a detector count rate of 100.000 cps was reached. The electrical output of both the built-in photodiode of the pulsed light source and the SNSPD detector were coupled to the individual channels of the oscilloscope.

The oscilloscope was set to measure the difference in arrival time between the electrical output pulses of the built-in photodiode and the detector, and store the data in a histogram over 100.000 detection events. The trigger levels were set to the steepest part of the rising edge of both the output pulses.

The timing jitter was determined by the full width at half maximum (FWHM) of a Gaussian curve fitted over the histogram. The jitter measurements were done at the operating current of each detector, unless mentioned otherwise.

Pulse amplitude

The pulse amplitude was measured by coupling the electrical output of the SNSPD detector to an oscilloscope (Lecroy, 4 GHz bandwidth and 40 GS/s sampling rate). The pulse amplitude was then determined by the maximum value of the 100 times averaged pulse. The reported pulse amplitude value for a detector is at a fixed operating current.